

Predicting airline acceptance of fuel-saving reroutes from flight-plan revisions

Ramon Dalmau^{a,*}, David Perez^b, Franck Ballerini^a, Seddik Belkoura^b,
Vincent Taverniers^b, Gratiel Marius Marin^b, Robin Deransy^a, Benjamin
Cramet^c, Grigori Gustin^c

^a*EUROCONTROL Innovation Hub, Brétigny-sur-Orge, France*

^b*EUROCONTROL Maastricht Upper Area Control Centre, Maastricht, the Netherlands*

^c*EUROCONTROL Headquarters, Brussels, Belgium*

Abstract

Every flight-plan revision records a choice an airline actually made: the route it dropped, and the route it filed instead. Eighteen months of European air traffic hold 1.48 million such pairs, read here as revealed preferences and used to train a ranking model that predicts, between two candidate routes, which one an airline would choose. Tested on data it never saw during training, the model identifies the chosen route 68.1 % of the time, rising to 94.9 % on the fifth of decisions it is most confident about. That confidence is calibrated, so a proposal channel built on the model can offer a cheaper route only when acceptance is likely, at a precision fixed in advance. Simulated against what the airlines actually filed next, such a channel reaches 79.2 % precision and surfaces tens of kilotonnes of planned-fuel savings a quarter. Its clearest blind spot is the flights that never revise at all, though a much scarcer set of matched pairs closes most of that gap. Airline route acceptance, this paper shows, is learnable from records the network already keeps.

Keywords: air traffic flow management, airline preferences, learning to rank, fuel efficiency

*Corresponding author.

Email address: `ramon.dalmau-codina@eurocontrol.int` (Ramon Dalmau)

1. Introduction

On a spring morning in 2026, an Airbus A320 was filed from Amsterdam to Barcelona on a route carrying 102 minutes of attributed air traffic flow management (ATFM) delay, and the airline subsequently re-filed. The new route shed that delay completely, and it cost the airline on each of the three counts an airline pays for directly: 39 minutes of flight time, 1 397 kg of extra planned fuel, and 466 euros more in en-route charges. Both routes are held in the network’s records and so, implicitly, is the airline’s judgement that the delay would have cost it more than the detour did.

That judgement is the quantity this paper models. A flight plan is not filed once and left alone: an airline may revise it at any time until the aircraft leaves, and the network’s response is only one of the reasons for doing so. Attributed delay of the kind the opening flight was carrying is the most visible prompt; an aircraft arriving late from its previous leg is another, since the delay it brings consumes the slack the rotation was holding, and an updated forecast is a third, since changed winds change the time and fuel every candidate route would need. A route may also simply cease to be fileable, in which case the revision is not a choice at all. What the archive records is what changed rather than why, and it does not separate one reason from another; every revision nonetheless leaves the same trace, the route replaced and the route that replaced it, and therefore a preference the airline expressed with its own filing system rather than in answer to a survey.

The scale of what is at stake is published every year. A single function coordinates air traffic flow across national borders in Europe, matching demand against declared capacity, and an independent commission reviews its performance annually, reporting how much further the traffic flies than a direct reference between the points at which it enters and leaves the airspace (Performance Review Commission, 2026); not all of that distance is avoidable, since airspace that cannot take all the traffic that wants it has to be protected, and a flight routed around it flies further, so part of the figure is what respecting a limited capacity costs. What this paper takes up is whether any of the remaining excess is a cheaper route the airline would have been willing to fly and did not file, an assumption made here rather than shown (Section 8 returns to it; a route closed by a restriction or an unavailable conditional route is not one of them, since it could never have been filed). Candidate routes are not hard to come by; what is hard is knowing

which candidate this particular airline would accept, on this day, with this rotation to protect, and knowing it while there is still time to file.

A proposal channel could fill part of that gap: a service that would offer an airline an alternative route that burns less fuel and that the model judges the airline would accept, inside the planning window and before the flight departs. The timing is not free: two of the model’s strongest inputs, the attributed delay and the airspace constraints behind it, exist only once a plan has been filed, so the channel would speak after a first filing rather than before one, and Section 8 returns to what that costs.

The objective of such a channel, and of this paper, is twofold: a route that burns less planned fuel emits less carbon dioxide when it is flown, but the saving must not be bought at the airline’s expense, since a route that saves a tonne of fuel while costing an hour of delay or a broken rotation is not a saving any airline will take. What this paper supplies is the instrument that lets both halves be pursued together, namely a way of ranking candidate routes by how likely the airline is to accept them, so that the environmentally cheaper proposals can be aimed at the flights on which they are also cheap in the airline’s own terms. Proposing routes that airlines reject is worse than proposing nothing at all, since a channel that is ignored is a channel that is eventually switched off.

A preliminary version of this work was presented at the SESAR Innovation Days 2026, where the question was whether route acceptance can be predicted at all. The present paper is the treatment of the channel: what it would address, at which operating points, for which airlines, and how the figure survives the design decisions behind it. It adds the baseline ladder and the calibration table behind those results, the channel simulation across four operating points, and the concentration of its value across operators and its sensitivity to the bias correction.

The contribution of this paper is threefold. First, it shows that airline route acceptance is learnable at network scale from revision events alone, and that the confidence of the resulting model is calibrated well enough to support a chosen operating point rather than a single accuracy figure. Secondly, it argues that the label construction is itself a design decision worth defending: a revision pair records a preference the airline actually expressed, in contrast with the usual alternative of ranking the filed route above candidates from a route generator that the airline may never have evaluated, a distinction Section 2.3 develops in full. Thirdly, it quantifies what a proposal channel built upon such a model could address, using the

airlines’ own subsequent filings as ground truth. That quantification also reports how the value concentrates across airlines and how sensitive it is to the design decisions.

The remainder of this manuscript is organised as follows. Section 2 explains why a revision constitutes a preference observation and why it carries a stronger label than generated candidates; Section 3 positions the work in the literature. Section 4 describes the archive and pair construction, Section 5 the model and its calibration, Section 6 predictive performance, and Section 7 the proposal-channel simulation. Section 8 collects the threats to validity, and Section 9 concludes.

2. Background

This section describes fundamental concepts that are not a contribution of this work, but that are needed to follow the sections that come after it. Thus, a reader who is already familiar with the European filing process and with discrete choice may move on to Section 3.

2.1. *Flight-plan revisions as preference data*

In the European network a flight plan is filed before departure and may be revised at any time until the flight leaves. Every filing is a message, and so is every later change, and the network keeps them all, so that what it holds for a flight is the whole sequence of messages sent for it. A route change is accordingly a pair of consecutive messages rather than a record of its own: the message that changed the horizontal route of the plan, and the message that preceded it.

The two messages of a pair together describe a decision, because the airline had committed to the earlier route and then filed the later one for the same flight on the same day, which makes the later route the one it chose and the earlier one the route it abandoned. That the pair is assembled from two messages sent at different times is not a detail of bookkeeping: it shapes the encoding, and Section 4.3 returns to it.

The sharpest of those decisions, and the ones in which the terms of the trade are clearest, are made under flow management. When the traffic expected in a volume of airspace exceeds what the controllers there can safely handle, whether because of demand, staffing, weather or equipment, a *regulation* is placed on that volume for a period, and the flights planning to cross it are given calculated take-off times at the rate the volume can absorb; a flight

held back this way is given a take-off time later than the one it requested, and the difference between the two is its *attributed ATFM delay*, the aircraft waiting on the ground rather than in the air. Not every flight crossing a regulated volume is delayed, since a lightly binding regulation absorbs most of its traffic at the requested times, and a flight caught by several regulations is counted against the one that penalises it most.

An airline facing such a delay has a choice that is genuinely economic: accept it, or file a route that avoids the constrained volume at the cost of distance, fuel, en-route charges and the departure slot it already holds, and neither answer is universally correct. For instance, a flight with a tight aircraft rotation (i.e., a schedule in which the same aircraft flies several legs in the day, so that delay on one leg propagates to the next) may find forty minutes of delay far more expensive than a tonne of fuel, whereas the same delay on the last leg of the day may well be cheaper than the detour. The decision is worth learning rather than deriving, since the trade depends on airline-specific and flight-specific costs that the network does not observe: published European reference values (Cook and Tanner, 2015) give the order of magnitude of the cost of delay and are widely used for analysis, but they are not the cost function any particular airline applies, and what an individual operator would charge itself for forty minutes is not known outside it.

Most revisions, though, are not a response to flow management at all: in seven pairs in ten neither route carries a regulation, and Section 4 reports what that majority looks like. A dispatcher re-planning after a forecast update, or after the aircraft’s earlier legs have moved, leaves the same kind of trace as one escaping a delay, and the archive records what changed rather than why.

2.2. Two operational use cases

The two uses the model is intended for share exactly one quantity, the probability that a particular airline would accept a particular alternative for a particular flight, and differ in almost everything else. The first is a proposal channel: for a flight whose filed route has a cheaper alternative, the channel proposes it only when the model judges the airline would accept, on the grounds that flight planning happens against a deadline and a cheaper, acceptable route can pass unnoticed simply because nobody looked at it in time. This is the use case Section 7 simulates, with the airlines’ own later filings standing in for the answers a real channel would collect. The second is releasing an overloaded volume of airspace, that is, removing

enough flights from it to bring demand back within capacity: existing formulations for ordering which flights to approach optimise a network objective without modelling whether the airline would fly what it is offered, while collaborative variants treat the airline’s preference as an input it must supply (Section 3). A calibrated acceptance probability supplies that input for every flight whether or not its airline has declared anything, changing the ordering so that one action relieves the overload, cuts fuel burnt, and asks least of the airline; acceptance stays voluntary, so this reorders a decision taken in any case rather than adding a new intervention.

Neither use case has been tested end to end: the first is simulated against what airlines subsequently filed, so the figures here are what a channel would have achieved had airlines behaved as they in fact did, and the second is enabled rather than demonstrated, since the model supplies the missing input but no overload-release experiment is run here. Both, if ever put into practice, would generate the labels they need, since every proposal answered is a new observation of the kind the model is trained on, but those answers would be a biased sample, since an airline can only answer proposals it is sent and those are the ones the model was already confident about; separating that bias from genuine acceptance is a question this analysis leaves open.

2.3. Revealed preference and the construction of pairs

Readers of this journal will recognise the setting as a route-choice problem, a family with a long history in travel-demand modelling (McFadden, 1974; Ben-Akiva and Lerman, 1985; Train, 2009). The observation unit here is the choice between exactly two alternatives that the decision maker itself constructed, which sidesteps the choice-set generation problem that route-choice models must otherwise solve, and the estimator is a gradient-boosted ranker rather than a parametric logit.

Roughly speaking, the model of this paper is a conditional logit whose utility function is learned nonparametrically. The training objective penalises the model according to how far the abandoned route’s score sits above the chosen one’s, and keeps pressing even once the order is right, exactly as a binary logit on utility differences would; what differs is that the score is an arbitrary function of the features rather than a linear index.

What the nonparametric score gives up is the economic interpretability of coefficients; what is gained is the ability to absorb the waypoint sequence itself, which Section 6.5 shows to be the largest single term in what moves the model, though smaller than the four cost indicators taken together. The

economic tradition supplies the framing that matters here, namely revealed rather than stated preference (Samuelson, 1938): every label in this paper is something an airline did, not something it said it would do.

Revealed preference has a direct consequence for how a training pair ought to be built, and it is the decision on which everything that follows rests: the natural way to build such a pair without revisions is to take the filed plan as preferred and a set of generated routes as less preferred. The authors’ own earlier work on ranking flight routes, whose purpose was to predict the flight plans not yet filed so that traffic demand could be estimated earlier than the filing horizon allows, was constructed in that way (Dalmau et al., 2024): its training pairs take the filed plan as preferred and a set of generated proposals (up to eleven route options produced by the Network Manager, the network-level flow-management function of EUROCONTROL) as less preferred, and on the pre-tactical set none of the latter need match the filed route at all.

A model can satisfy a label built that way by learning what a filed plan looks like rather than what an airline wants: a generated candidate may join published segments in a way no dispatcher would ever file, so that telling it from a filed plan can be done on the shape of the route alone, without knowing anything about the airline that would fly it, and recognising filed plans and ranking routes by preference are different skills that an accuracy figure cannot distinguish. A revision pair carries no such doubt, and Fig. 1 draws the two constructions side by side on one city pair: both routes of a revision reached the dispatcher’s screen, since the airline filed each of them in turn, for the same flight, on the same day, so the route labelled less preferred is not an alternative constructed after the fact but one the airline had committed to and then dropped, and whatever else was happening, both routes were under the airline’s own consideration. No dataset offers both label sources for the same flights, so this is an argument about how the labels are built rather than a measured comparison.

The question also narrows, in that the model is not asked what an airline will file but which of two routes it prefers, a strictly easier question and precisely the one a proposal channel needs answered; the pair, though, is the unit of *training* and not a limit on use, since the trained model gives a score to one route at a time against the other candidates on offer for the same flight (the encoding of Section 4.3 expresses each indicator as the amount by which that route exceeds the cheapest candidate, so what the model needs is the candidate set, not a pair): a flight offered five routes is scored five times against the cheapest of the five, and one offered fifty is scored fifty times,

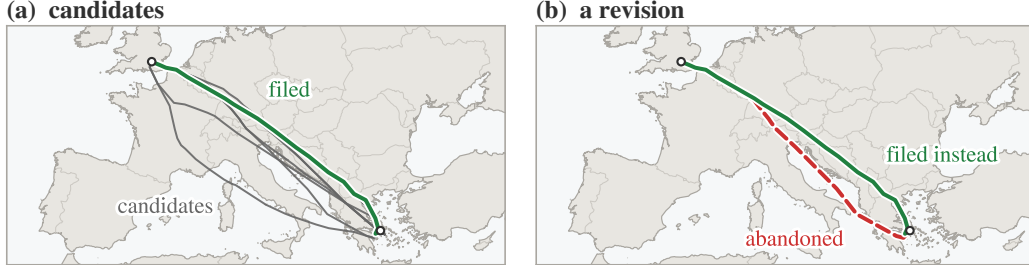


Figure 1: Two ways to build a training pair, on one city pair; each route is labelled by its role where it is drawn. Panel (a): the filed route (solid green) is ranked above candidates from a route generator (grey), which the airline may never have evaluated; those drawn here are other routes observed on the same city pair. Panel (b): a revision supplies two routes that the same airline filed for the same flight hours apart, one of which it abandoned. Only the right-hand pair records a preference the airline actually expressed.

with the pairwise objective doing nothing more than teaching the model where to place those scores relative to one another. Every figure in this paper is nonetheless measured on pairs, because pairs are what the archive records.

The price to be paid is that revisions are not a random sample of flights, but rather flights that changed, which introduces a bias of its own. Rather than assuming that bias away, Section 6.3 measures it.

3. Related work

Three threads of literature meet in this paper, and the first, rerouting in air traffic flow management, has largely approached the problem from the network side. Rerouting was formulated as a dynamic network flow problem (Bertsimas and Patterson, 2000) and later as a stochastic extension that reroutes flights as capacity evolves (Mukherjee and Hansen, 2009); both optimise a network objective and assume the resulting assignment is executed, so the airline’s willingness to fly it is not modelled.

The value of airline collaboration in flow management has been examined (Castelli et al., 2012), which motivates the channel simulated here but not which proposals would be taken up. Closer still, incorporating airline preferences into collaborative rerouting changes which reroutes are selected (Murça, 2018; Xu et al., 2020), as does the collaborative trajectory options programme, where an airline attaches to each alternative a cost relative to

its preferred option (Kulkarni, 2018). In all of these the preference is an input the airline must supply; this paper learns the preference function itself, at network scale, from what airlines did rather than stated, supplying that input for every flight whether or not its airline has declared anything.

Preferences have also been estimated rather than collected: trajectory options derived from historical traffic by clustering and classification feed a demand-capacity optimisation model (De Giovanni et al., 2024), and the probability that a regulated flight reroutes of its own accord has been predicted so a flow manager can weigh a regulation’s consequences before activating it (Dalmau, 2022), though both estimate what airlines tend to do absent a proposal, whereas releasing an overloaded volume needs to know whether a particular flight would accept a particular alternative, the quantity this model supplies. Beyond the research literature, airlines already express flight-level priorities within a regulation through the user-driven prioritisation process (Pilon et al., 2021; Ruiz et al., 2019), which orders delay among an airline’s own flights rather than addressing route choice, and would complement a preference model of this kind rather than replace it.

The second thread is economic: earlier than any of the above, route choice in Europe was studied as a trade between fuel and en-route charges (Delgado, 2015), establishing that the choice is economic and learnable but using no label in which the airline rejected an alternative it had itself filed, and the discrete-choice tradition of Section 2.3 supplies the theoretical frame for such labels; what it has lacked in this domain is a data source in which both alternatives were genuinely available, which revisions supply.

The third thread is methodological: learning a ranking function from pairs is standard (Burgess et al., 2005; Liu, 2009), and learning preferences from logged behaviour rather than elicited judgements was established in information retrieval, where clicks on search results yield reliable relative, though not absolute, relevance labels (Joachims, 2002). The revision archive is the aviation analogue of a click log, in that nobody was asked anything and each record is trustworthy only as a comparison between alternatives actually available; the same literature supplies the caution this paper inherits, that logged data reflects the behaviour of the system that logged it (Section 8). The direct predecessor of this work (Dalmau et al., 2024) established the framing adopted here, a gradient-boosted ranker scoring routes by their indicators in context, and reported the difficulty motivating the present paper: its training pairs rank the filed plan above candidates the airline may never

have examined, the argument and label construction Section 2.3 sets out in full.

4. Data and pair construction

This section covers the archive, the pairs built from it, and the encoding under which they reach the model; the model and the calibration follow in Section 5.

4.1. *The archive*

Eighteen consecutive months of European network data, from January 2025 to June 2026, are used. After retaining only those revisions that change the horizontal route, and only those pairs with complete key performance indicators, the archive contains 1.48 million labelled pairs. Each route is described by twenty-three features, summarised in Table 1, one of which is in-house shorthand and worth spelling out: the protected volume is the volume of airspace whose capacity the regulation was declared to protect.

“Airline” is used loosely throughout for any aircraft operator, since the archive contains business, cargo and non-scheduled traffic as well as scheduled airlines, and the figures say “operator” where the count is over all of them. All the data used is held by the network for operational purposes and was processed under existing data-handling arrangements. Operator identity enters the model as a categorical feature only, and no operator or airspace volume is named anywhere in this paper. A small number of individual flights are reported as worked examples, and nothing beyond what each example needs is given about them.

What those 1.48 million decisions look like matters before any model touches them, and Table 2 breaks the archive down by the regulation context in which each revision was made.

Seven revisions in ten carry no regulation on either route, so most of the archive records ordinary flight-plan housekeeping rather than escapes from regulations, and the median fuel difference within those pairs is a single kilogramme. Where a revision does escape a regulation it pays for the escape, since only 28.1 % of escape revisions move to a route that burns less fuel and the median move costs 101 kg and 2.1 minutes.

The mirror image of an escape is the smallest context of all, the 1.4 % of revisions that took on delay, which is the only context in this eighteen-month table in which the share of fuel-saving moves exceeds one half, because

Table 1: What the model sees for each route; some rows summarise several features (twenty-three in all). The first two groups and the last belong to the route and differ within a pair, the indicators entering as differences from the pair minimum; the rotation and context rows belong to the flight and carry the same value on both routes, the one exception being the outbound connection time, which is shortened on the slower route by the difference in flight time and is therefore route-dependent after all.

group	content
indicators	flight time, distance, planned fuel, en-route charges (the fees an airspace user pays for crossing each charging zone)
regulation	attributed delay (own, inbound, outbound), their count, reason, and the type and identifier of the protected volume
rotation	inbound and outbound connection times and states
context	airline, aircraft type, city pair, hour, weekday, month, time to departure
route	the ordered waypoint sequence, as text

Table 2: The revision archive by decision context, over all eighteen months. A pair is regulated here when a regulation was attributed to the flight while filed on that route (the delay itself is free to be zero; Table 4 sorts the held-out quarter by the delay instead, giving different counts). Unregulated pairs carry no regulation on either route, escapes carry one on the abandoned route only, the next two carry one on both, and the last only on the newly filed route; the median fuel difference is between the two routes of a pair, and the final column is the share where the newly filed route burns less.

context	pairs	share [%]	median Δ fuel (new – old) [kg]	new route saves fuel [%]
unregulated	1 047 228	70.6	1	46.2
escaped	117 594	7.9	101	28.1
reduced delay	155 326	10.5	65	31.6
kept or raised delay	141 648	9.6	0	47.4
took on delay	21 337	1.4	–5	51.8

a flight that accepts delay tends to be paid in fuel for it. Everywhere else the share stays below one half, which already suggests that fuel is not what filing behaviour optimises, an observation that Section 6.1 sharpens into a measured result. Figure 2 shows two revisions drawn from the archive, an escape and a fuel saver.

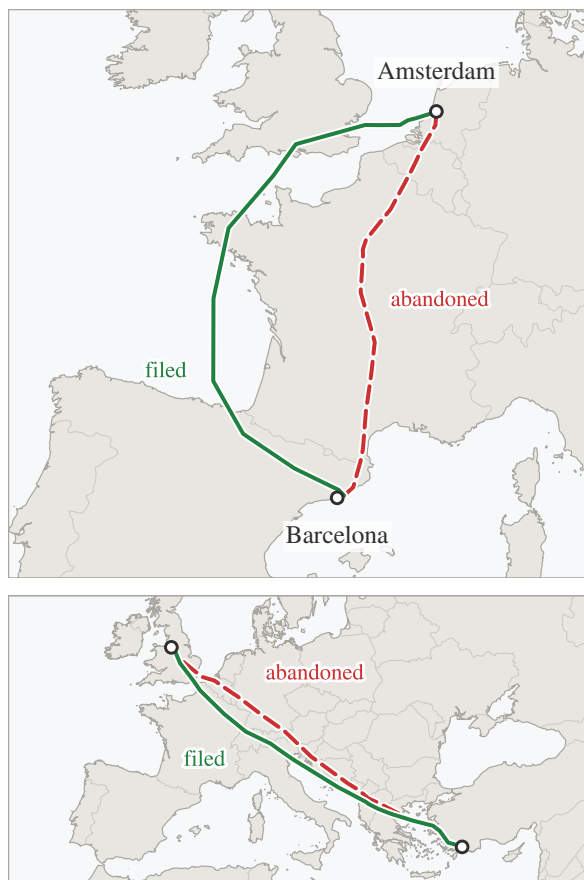


Figure 2: Two revisions from the archive, in the visual language used throughout this paper: the abandoned route is dashed red and its replacement solid green, each labelled on the track itself. Top: an escape, in which the airline paid fuel and time to leave a regulated volume; the volume itself is not drawn, since naming the airspace would identify it, and the two airports are named because this city pair is the running example of Section 1. Bottom: a fuel saver, in which the replacement was cheaper outright; its endpoints are left unnamed, that running example being the only city pair this paper identifies. The first kind accounts for a quarter of the regulated archive; the second is what a channel of this kind exists to surface earlier.

4.2. Stay pairs

Every revision pair is, by construction, a flight that *did* change route, and that creates a trap. A model trained only on such events learns that change is what happens, and recommends it far too eagerly against traffic that mostly stays, by a margin that Section 6.3 measures. Put another way, the archive

answers one question only, namely: when an airline changed its route, which route did it move to? It cannot answer the complementary question, because a flight that kept its route generates no record of the alternative that it declined.

A second, much smaller dataset was therefore constructed, in which *keeping* the route is the revealed choice. Let us consider two flights on the same day, with the same origin, destination and aircraft type, that flew different routes and neither of which revised its plan, and let us suppose that one of them was caught by a regulation while the other was not. The regulated flight had a reason to move and an alternative that was demonstrably flyable that day, namely the route its twin was flying, and yet it stayed; its route is accordingly labelled as preferred, and that of its twin as the alternative it declined.

What the twin does not establish is that the same routing was open to the stayed flight at its own departure hour, since route availability in Europe is conditioned on the time of day, and that limit sits alongside the regulation confound described next.

The matching conditions are restrictive and the yield is correspondingly low: 6 672 such *stay pairs* (pairs in which the preferred route is the one that was kept rather than the one that replaced it), of which 1 093 fall in the test months, against 1.48 million revision pairs. The two are used together, with stay pairs entering the fitting set at an increased group weight (the pair's weight in the ranking loss).

The stay pairs also carry evidence of their own, and it concerns the price airlines put on staying where they are. The flight that stayed faced a median peak delay (the largest delay attributed to it at any point before departure) of 31 minutes, and in 47.5 % of the pairs the route it kept burns *more* planned fuel than the alternative its twin was flying (the median difference is 3 kg in the kept route's favour, i.e. essentially nil). Airlines therefore accept delay even when a demonstrated, comparably priced alternative exists, which is revealed preference for stability itself, and precisely the behaviour an over-eager proposal channel would violate.

One property of this construction has to be stated alongside it, because it bounds what the stay pairs can ever be used to measure. The regulated flight is always the one whose route was kept, so regulation status identifies the labelled answer. Concretely, in all 1 093 stay pairs of the test months the kept route carries a positive attributed delay and at least one regulation, and the alternative carries neither, with no exceptions. A rule as crude as

“prefer the route of the flight that was regulated” is therefore right on every stay pair without examining a route at all. Whether the model takes that shortcut is a question the archive can answer, because it also contains the mirror population in which the same rule is always wrong; Section 6.3 puts the two side by side.

4.3. Encoding and leakage control

Treating a revision as a labelled preference pair requires care, because the two routes are separated in time as well as in geography: the replacement is by construction the later message, and any feature that drifts with time can therefore identify the answer without saying anything about the quality of the route. On this archive this is not a technicality: a model built without attention to it would report excellent figures while knowing nothing whatever about routes.

One pair from the held-out quarter shows the size of the problem. Its original plan was filed at 00:30 and the plan that replaced it at 10:56, for a flight due off the blocks at about 12:30; left as filed, the time remaining to departure would read 720 minutes on the abandoned route against 94 on the one that replaced it. That gap is not an outlier in kind: across the held-out quarter the median distance between the two filings is 33 minutes, a quarter of the pairs are more than an hour and a half apart, and the revision is the later message in all but a thousandth of them. A model given the feature in raw form could therefore answer “the one filed later” and be right on essentially every pair while knowing nothing whatever about routes, leakage rather than learning that would be invisible in the accuracy figure, since the model would score close to perfectly on held-out revisions for the empty reason that they carry the same timestamp asymmetry as the training ones.

To remove the shortcut, all flight-level attributes were harmonised across the pair so that only route-level quantities differ, and the filing-time feature was collapsed to the pair minimum, giving it an identical value on both routes; the stay pairs of Section 4.2 pass through the same step, the values of the flight that stayed copied to the other row, so that in both populations only route-level quantities separate the two candidates. A subtler shortcut hides in the absolute indicator values: on a 400 km flight where the two routes burn 2 000 and 2 300 kg, a model shown the raw magnitudes learns, inevitably, what 2 000 kg means on this kind of sector, and with it everything that correlates with sector length (which airlines fly it, at which hours, with which aircraft), a shortcut that generalises poorly to sectors of other lengths.

The within-pair encoding subtracts the pair minimum from each indicator, so what reaches the model is 0 against 300 kg, a pure comparison carrying the same meaning whether the flight is 400 or 4 000 km.

4.4. *Splits*

The evaluation is strictly out of time, in the sense that the model is fitted on January 2025 through February 2026 (1 134 064 pairs), tuned on March 2026 (67 349), and evaluated once on April to June 2026 (281 720), the held-out quarter cited throughout; the three account for all 1.48 million pairs of the archive. Every threshold, calibrator and stopping decision is taken on the tuning month alone, flights appearing in the tuning or evaluation months are removed from the fitting set, and eleven automated checks verify split disjointness, feature construction and the pair invariants.

5. **Model, training and calibration**

This section covers the scoring model, the objective it is trained under, and the calibration step that turns its scores into acceptance probabilities. The design follows the predecessor system (Dalmau et al., 2024) except where stated, and the emphasis here is on the two decisions that the results show to matter, namely the within-pair encoding and the stay-pair mixture.

The operational question is comparative: given two routes for one flight, which does the airline prefer? The model gives each route a single number, not a probability and with no units, meaningful only next to the number given to another route for the same flight. Training pushes the two scores apart in the right direction: for every historical pair the model pays a penalty, largest when the abandoned route outscores the chosen one, falling as the chosen route’s lead grows but never quite reaching zero, so the model keeps pressing even once the order is right (Burgess et al., 2005). Nothing in the objective asks the model to reproduce a route in isolation, which is what makes the pair the natural unit and connects the estimator to the conditional-logit tradition of Section 2.3.

Gradient-boosted decision trees (Prokhorenkova et al., 2017) are used as the scoring function, at a depth of 8, with the learning rate selected automatically and early stopping on the tuning month. The waypoint sequence is carried as text and encoded as a bag of tokens, each waypoint identifier treated as a word and the route as the unordered collection of its words; order is discarded, and what survives, which published points a route uses,

is already enough to let the model exploit route structure rather than aggregate distance alone. For instance, two routes on the same city pair may differ by under one per cent in distance and in planned fuel, indistinguishable on every indicator, while only one crosses a volume whose identifier the model has met in thousands of earlier revisions; the token set carries that fact and the aggregate distance cannot.

Stay pairs enter the fitting set with a group weight w , and the weight is swept over the values 10, 30 and 100; the selection of $w = 10$ is discussed with the change-bias results in Section 6.3.

Once trained, the model has still to be connected to a decision, and a raw score is not sufficient for that. On a real pair from the held-out quarter, the model scores the two routes -2.82 and $+0.90$; neither number means anything on its own, since the score has no units and the same two numbers on another flight would carry no comparable meaning, but their difference, 3.7 in favour of the second route, is not yet a decision either, because nobody in an operations room knows what a gap of 3.7 is worth. Calibration is the step that turns the gap into a statement of the form “about eight times in ten, the route the model prefers is the one the airline goes on to file”; the conversion is a monotone mapping from score gap to observed rate, fitted on the tuning month alone (Zadrozny and Elkan, 2002) and applied unchanged to the evaluation months. A gap of half a point maps to a probability of about 0.72 , against which the higher-scoring route in that band was the one the airline filed in 68.5% of held-out cases; the 3.7 of the example above maps to 0.99 , and in that band the model was right 97.4% of the time. The stated probability tracks the observed share to within 4.4 points at worst, and 1.8 on average across fifteen finer bands, mild over-confidence through the middle of the range.

It is the probability, never the gap, that the proposal policy of Section 7 compares against its threshold, and it is calibration that makes a precision target meaningful at all, since without it one can rank proposals but cannot decide how many of them to make.

Three mappings were compared on the tuning month, namely the raw sigmoid of the score gap, Platt scaling (a two-parameter sigmoid fit) and isotonic regression (a stepwise monotone fit), of which isotonic regression reached the lowest tuning-month log loss and is what the pipeline uses. Miscalibration is common in flexible models fitted this way (Guo et al., 2017), and this model runs the other way on a different criterion from the one that chose the calibrator, in that on the held-out quarter its own raw sigmoid was

better calibrated than the isotonic fit selected to correct it: the expected calibration error (the mean gap between stated probability and observed rate) is 0.9 percentage points for the sigmoid against 1.8 for the isotonic fit. The selection was made on the tuning month before the evaluation months were touched and is reported unrevised. The reading the authors draw from it is that on this archive the fitted mapping is a safeguard rather than a correction, and that using the sigmoid instead would be a reasonable choice in practice.

6. Predictive performance

This section reports how well the model predicts, and then decomposes that figure. Accuracy is set first against coverage and against the simple rules a channel could use instead, and then resolved into the strata where it lives, the design choices that earn it, the errors that remain and the features the model turns out to weigh.

6.1. Accuracy, coverage and baselines

On the held-out quarter the model identifies the airline’s chosen route in 68.1 % of 281 720 pairs (95 % interval 67.9–68.3, or 67.7–68.6 under city-pair clustering), against the 50 % of random choice. Reassuringly, the tuning month scores 67.5 %, and there is therefore no sign of overfitting to it.

Accuracy alone understates operational value, because the model is informative about its own uncertainty. Fig. 3 reports accuracy when the model is permitted to abstain on those cases in which the two routes score close together. Let us call the share of flights that the model is allowed to answer its coverage, the remaining flights being left alone. On its most confident fifth of the cases the model is right 94.9 % of the time, and a proposal channel is not obliged to answer every flight: it operates at the left of this curve.

The curve is one reference; the other is what simpler methods achieve on the same held-out quarter. Table 3 sets out the whole ladder, from a coin toss to the model of this paper, so that a reader can see where the accuracy is bought and judge whether the complexity earns its place.

No single cost indicator predicts the choice well: airlines weigh delay, flight time, fuel and route charges against one another, and no one quantity dominates. The structural reason is set out in Table 4 below: most revisions buy something, usually relief from delay, and pay for it in fuel and time, so a route’s cost in isolation says little about whether it is the one the airline

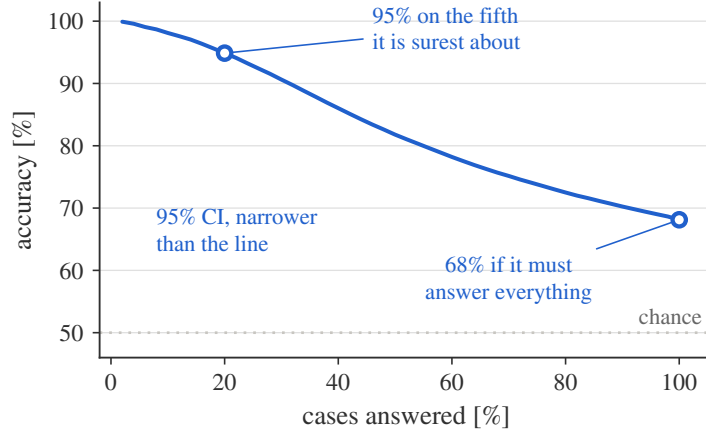


Figure 3: Accuracy against coverage on the held-out quarter. The horizontal axis is the share of cases the model is required to answer, lowest confidence dropped first; the vertical axis is the share of answered cases it gets right. Moving left, the model answers fewer cases and is right more often: 95 % of the fifth it is surest about, against 68 % when it must answer everything. A “case” here is one pair of routes, and answering it means naming one of the two. The dotted line is chance; the 95 % confidence interval is drawn but is narrower than the line itself at every point. It is this curve, rather than any single figure on it, that characterises the system.

chose. A proposal system built on any single cost indicator and nothing else would therefore struggle to track the airline’s own subsequent behaviour on most revised flights, which is exactly why Section 7 builds its policy on the learned model rather than on any one indicator.

Beyond the rules shown in Table 3, the strongest single-feature rule tested on this archive reaches 55.0 %, which puts the model’s margin over the best rule built on any one indicator at thirteen points rather than fifteen. The delay rule remains the instructive one, and the one shown in the table: it is right 71.6 % of the time, but only ever decides on the 14.6 % of pairs where delay differs at all, and scoring the rest as coin tosses gives $0.146 \times 71.6 + 0.854 \times 50 = 53.2\%$. Delay avoidance is real, the model must and does learn it, and it accounts for only a small share of the archive.

The learned baselines settle the remaining question, namely whether the additional complexity is doing any work. A logistic model given the four cost indicators as within-pair differences reaches 54.9 %, and the attributed delay and the count of regulations take it to 56.4 %, where eight further numeric features leave it unchanged. Adding the categorical context takes it

Table 3: Baselines on the held-out quarter, 281 720 pairs; not every single-feature rule that was tested is shown, only those that also anchor later discussion. A single-feature rule decides only where both routes carry a value for its indicator and the two values differ, and is scored as a coin toss on the rest; the “decides” column gives that share. A route under no regulation carries no delay figure rather than a zero, which is why the delay rule decides on fewer pairs than the share on which the revision changed the delay in Table 4. The logistic models are fitted on the same 1 134 000 training pairs as the model of this paper, with the same within-pair encoding, and their regularisation strength is selected on the tuning month.

baseline	inputs	decides [%]	accuracy [%]
coin toss	0	–	50.0
prefer less attributed delay	1	14.6	53.2
logistic, cost indicators	4	–	54.9
logistic, and delay and count	6	–	56.4
logistic, all numeric	14	–	56.4
logistic, and categorical context	22	–	57.5
this paper’s model	23	–	68.1

to 57.5 %, at which point it has seen every input the model of this paper sees except the waypoint sequence. The gradient-boosted model reaches 68.1 % on those same inputs plus the route text, so roughly eleven points come from the nonlinearity and the route representation together rather than from the choice of features.

That margin establishes that route choice carries structure that neither a cost rule nor a linear index in the same inputs recovers, which was by no means self-evident, but it does not establish that 68.1 % is anywhere near the ceiling. The residual splits in two. The first part is not information at all: the two routes of a pair are often nearly the same route, and the model scores 61.9 % where they share more than nine tenths of their waypoints against 74.9 % where they share less than a quarter, so a pair differing by a single kilogramme of planned fuel may give the airline no reason to prefer either. The second part is a property of the decision rather than of the archive: a route is chosen by a person, and by different people on different days, from information no route-level record holds (how much duty time the crew has left, a swap of departure slots between the airline’s own flights, a company routing policy, a weather judgement not yet materialised), so

Table 4: What each revision did to the flight’s attributed ATFM delay, on the held-out quarter, and how well the model reads it. Fuel and time differences are the new route minus the one it replaced, so a positive number is a cost the airline accepted; intervals are Wilson 95 %. This table is keyed on the delay itself, whereas Table 2 is keyed on whether a regulation was attributed to a route, and a regulation can be attributed while leaving the delay at zero, which is why 71.0 % of this quarter’s pairs touch no delay against only 63.3 % carrying no regulation (stated with the regulation-exposure strata, below).

what the revision did	pairs	share [%]	median Δ fuel [kg]	new saves fuel [%]	accuracy [%]
no delay either route	199 932	71.0	0	47.7 64.0	[63.8, 64.2]
removed the delay	35 890	12.7	103	26.8 88.2	[87.8, 88.5]
reduced the delay	21 202	7.5	56	31.8 81.5	[81.0, 82.1]
left the delay alone	10 234	3.6	1	45.5 60.4	[59.5, 61.4]
increased the delay	9 189	3.3	−2	52.2 61.0	[60.0, 62.0]
took delay on	5 273	1.9	−10	53.2 60.9	[59.5, 62.2]

two competent flight planners facing the same situation may reasonably file differently without either making a mistake. Perfect prediction would require route choice to be a deterministic function of what the network records, and there is no reason to expect that it is.

6.2. Accuracy by stratum

The 68.1 % headline is an average over decisions of very different kinds, and the first question a reader should ask is what kinds of decision the archive actually contains. Table 4 answers that and the performance question together, on one population: it sorts the held-out quarter by what the revision did to the flight’s attributed ATFM delay, and reports how often the model called each sort correctly.

Seven revisions in ten touch no delay on either route, and the median such revision moves the flight to a route costing the same planned fuel to the kilogramme: this is ordinary flight-plan housekeeping, and the model reads it at 64.0 %, which is well above chance and far below what it reaches on the delay strata.

The two rows in which the airline bought relief from delay are the ones the model reads well. When a revision removes the attributed delay outright, one revision in eight, the model identifies the chosen route 88.2 % of the time, and it is worth noticing what that relief cost: the median such move burns 103 kg more fuel, and only 26.8 % of these moves are cheaper on fuel at all.

Reducing rather than removing the delay costs a median 56 kg and is read at 81.5 %. The three remaining rows, in which the delay stayed put, grew, or was newly taken on, together account for 8.8 % of the quarter and are read at about 61 %.

The shape of that answer is what an operational reader most needs, and it divides cleanly in two. The model is good, at 81 to 88 %, precisely where an airline is trading fuel against delay and the stake is large, and weak, at 60 to 64 %, precisely where the two routes cost almost the same and the choice between them carries little consequence either way. That is a defensible division of labour in general rather than a defect, since it is the first kind of decision that a proposal channel is built to reason about. It is also, however, exactly the wrong way round for the channel of Section 7, which proposes on fuel and therefore speaks precisely where the model is weakest, and Section 7 takes that tension up.

One consequence of the archive’s regulation strata is worth stating before the headline is read anywhere else. The held-out quarter is not a random slice of the archive: 63.3 % of it carries no regulation on either route against 70.6 % of the eighteen months, and that unregulated stratum, where accuracy is only 64.0 % (against 88.0 % when only the old route is regulated, 70.6 % when both are, and 58.6 % when only the new one is), is the hardest of the four. Reweighting the four regulation strata to the archive’s own mix gives 67.1 % against the 68.1 % headline, so about one point of the headline is the composition of the quarter rather than the skill of the model.

6.3. Ablation and change bias

Table 5 reports screening ablations, each configuration removed in turn and the model retrained, and one of the five dominates the rest.

Expressing the indicators relative to the other route in the pair is decisively the most important choice, since removing it costs nearly ten percentage points: it converts four absolute magnitudes into a comparison. The waypoint-text result requires care and illustrates why screening ablations can mislead: at screening scale (the reduced setting of Table 5) the text feature appears worthless, yet retrained at full scale it is worth +1.4 percentage points (68.2 against 66.8, the 68.2 being a separate run of the adopted configuration, whose seeds span 0.17 points of accuracy), because a bag-of-words representation spread over thousands of rare tokens cannot repay its cost in 800 shallow rounds. It is a real but secondary contributor, worth about a

Table 5: Screening ablations, each variant retrained from scratch at a deliberately reduced setting (depth 6, 800 boosting rounds; 60.6 % on revision pairs). Values are the change in revision accuracy against that 60.6 % baseline, not the 68.1 % headline model; positive means the variant scores *higher* at that reduced setting. Differences are significant under McNemar’s test at $p < 0.001$ except the waypoint text at screening scale ($p = 0.58$).

variant	Δ accuracy [pp]
without within-pair KPI normalisation	−10.0
without stay-pair correction	+3.7
with monotone constraint	−0.2
without waypoint text	+0.03
including vertical-only revisions	−2.0

tenth of the model’s margin over the best single rule of Section 6.1, yet attribution ranks the same feature first, an apparent tension that Section 6.5 resolves.

A more principled alternative to the bag of tokens was tried before it was settled on: routes recast as scale-, rotation- and translation-invariant segment descriptors (each leg’s bearing change relative to the great circle, plus its length as a fraction of the whole), fed to a neural encoder trained under the same pairwise objective. It did worse alone than the tokens and added only a fraction of a point alongside them, too little to repay a second model in the serving path; a linear classifier on the encoder’s own representation separated preferred from abandoned routes barely better than chance, pointing at the geometric input rather than the objective as the limitation, likely because the token representation carries which published structure a route uses, not its geometry, and two geometrically similar routes through different structures are not interchangeable to an airline. These trials predate an archive repair that had dropped about a quarter of the valid pairs, so only their ordering, not their absolute accuracy, is claimed.

The change bias is measured on the held-out stay pairs, and one thing has to be settled before the figure can be read: Section 4.2 established that regulation status identifies the answer on every stay pair, so a model could in principle score well on them by reading the regulation and nothing else. The archive answers whether this one does directly, since it holds a second population with the same regulation signature and the opposite label: the escapes, where the abandoned route is the regulated one and the right answer is to prefer the unregulated route, against a stay pair, where the kept route

is the regulated one and the right answer is to prefer it. A rule reading regulation status alone must therefore score 100 % on one population and zero on the other.

The model scores 88.0 % on the 30 953 escapes of the held-out quarter and 90.0 % on its 1 093 stay pairs, at a cost of 0.8 points of revision accuracy (68.1 % against 68.9 % for a model trained without stay pairs); no reading of regulation status produces that pair of numbers, since telling the two situations apart requires the route and the flight around it, precisely what the stay pairs were added to instil. Trained on revisions alone, the model prefers the alternative route in 65.2 % of the cases where the flight demonstrably kept its own, carrying the lesson that a regulated route is one to escape into a population where the opposite holds; adding stay pairs at group weight 10 turns that 34.8 % right into 90.0 %, at the same 0.8-point cost (the 3.7 points of Table 5 are measured at the reduced screening setting, not at full scale).

Two bounds on that result remain, and neither is repaired by how it is measured: the stay set is small (1 093 test pairs against 281 720 revisions, putting the interval around 90.0 % at [88.1, 91.7]), and its two routes belong to two different flights, which no revision pair does, so a model could learn to recognise the construction itself rather than the preference it encodes. The balanced stay set of Section 8 would settle both bounds properly.

The weight itself was chosen by a sweep over $\{10, 30, 100\}$, one model trained per value: raising it from 10 to 100 buys 7.3 further points of stay accuracy and costs 1.5 points on revisions, a poor trade, so $w = 10$ is used because it takes nearly all of the available correction at the smallest cost.

6.4. *Error analysis*

Fortunately, errors concentrate where the model is already unsure. Only 5.6 % of them fall among the quarter of pairs whose two routes score furthest apart, against the 25 % that would fall there if error were independent of confidence. The same pattern appears from the other end, in that 27.7 % of the errors are near-ties, in the sense that the two scores land within one fourth of the archive’s median gap. Errors therefore concentrate at low confidence, which is the property that makes confidence filtering work.

The confident errors are nevertheless instructive. In one case from the held-out quarter, every cost indicator and the attributed delay favoured the route the airline went on to file, yet the model confidently scored the abandoned route higher; since flight-level context is identical within a pair by construction, and every route-level quantity but the waypoint sequence favoured

the filed route, the model’s preference for the abandoned one came from that sequence, presumably a learned structural preference that overrode the rest. Confident errors are therefore not all of the hidden-information kind, meaning decisions driven by facts that appear in no route-level record, such as a slot swap or a curfew.

Opportunely, such cases cost less than their share of the error rate would suggest, since a proposal channel would have said nothing here: the alternative it would have offered was the one already filed. The expensive errors are the opposite ones, in which the model confidently proposes a route the airline declines, and the change-bias measurement above is the closest available proxy for their rate.

6.5. Feature attribution

A ranking model that cannot be interrogated is of little use to an operational user who is asked to trust it. Shapley attributions were therefore computed over the held-out quarter: each feature is credited with the share of the score that is attributable to it, averaged over every order in which the features could have been supplied to the model (Lundberg and Lee, 2017). Because the model decides on a comparison rather than on one route, what is reported here is the *difference* between a feature’s attribution on the two routes of a pair, averaged in magnitude over the quarter (Fig. 4). That is a ranking not of what correlates with acceptance, but of what moves this particular model’s decisions.

The waypoint sequence is the largest single term in that ranking, at 0.40 against 0.26 for attributed delay and 0.20 for planned fuel, so route structure carries information about the choice that no individual cost indicator carries; the claim should be made at exactly that width and no wider, since the four cost indicators taken together sum to 0.52 and therefore outweigh the route text, and what the attribution supports is that route structure is information a cost model does not have, not that it dominates the decision. Attributed ATFM delay comes next, moving the score more than planned fuel does, though why it does is not something these figures settle, since the ranking is of what moves this model and a tree ensemble responds to how a feature is distributed rather than to the units it is measured in. What can be said is that nothing here shows airlines find delay the easier of the two to price; if anything the reverse holds, since fuel has a market price and the cost of delay is particular to the airline, which is exactly why the trade between them has to be learned rather than derived.

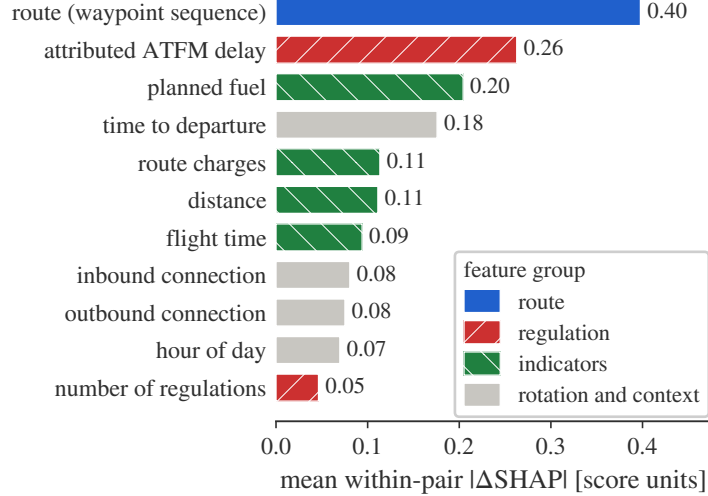


Figure 4: What moves a decision: the mean absolute *difference* between a feature’s Shapley value on the two routes of a pair, over the held-out quarter, in units of the ranking score, with each bar’s value printed beside it. The eleven features that move the score most are shown, of the twenty-three the model sees; fill colour and hatching group them, with the rotation and context groups of Table 1 drawn together. The waypoint sequence is the largest single term, ahead of every individual cost indicator, though the cost indicators taken together sum to more. A large attribution is not the same as a large ablation cost: removing the waypoint sequence and retraining costs 1.4 points of accuracy. The two measure different things, as the text below explains.

One entry in the ranking is a caution rather than a finding: the time remaining to departure appears fourth, at 0.18, even though the encoding of Section 4.3 makes it identical on both routes of every pair, so it cannot by itself separate them, and it does not. What the attribution measures here is its interaction with the features that do differ, so the same fuel difference is read differently three hours before departure than twenty minutes before it, a real effect rather than a residue of the shortcut the encoding removed, worth distinguishing because the two look alike in the chart.

Direction, not only magnitude, distinguishes the two leading terms. Attributed delay is penalised sharply, falling to slightly negative when the filed route is the more delayed of the two, and reaching 2.3 points in the filed route’s favour where it carried an hour and three quarters less delay than the one it dropped; planned fuel runs the other way, weakly, contributing a few

tenths of a point either way, in a sign that is a property of which flights revise (Section 6.1) rather than a general preference. Their comparable mean attribution magnitudes, 0.26 against 0.20, conceal how differently the two behave: fuel moves the score by a few tenths of a point on almost every pair, while delay moves it further but only on the minority of pairs where one exists at all.

Descending to individual waypoints sharpens the picture, since deleting a single identifier from a route and rescoreing shifts the score by up to 2.5 points for the most influential one, and within a route the points that carry that weight are a few of them rather than all of them evenly, which is what a model reading route structure would show and not what a model reading route length would show (which waypoints those are is not reported, since naming them would identify the airspace and, through it, the operators that use it). The waypoint text ranks first by attribution, yet removing it and retraining costs only 1.4 points of accuracy: attribution measures how much a feature moves the score of the model we have, ablation measures how much worse a model without it would be, and where features are correlated, as route geometry and cost indicators plainly are, the two answer different questions and the second is the one that governs what may safely be dropped.

7. Proposal channel

What a proposal channel is worth turns on how much fuel is at stake and on how confident one can be that the fuel is real. The simulation of this section answers both questions by refusing to count anything that the airline did not subsequently do.

7.1. Policy and operating points

For each pair the channel identifies the route that burns less planned fuel, and proposes it when the calibrated probability that *that* route is the one the airline will file exceeds a threshold τ . The distinction matters, because the cheaper route is often the lower-scoring one. The calibrator maps a score gap to the probability for the higher-scoring route, so when the cheaper route is the lower-scoring one the channel compares one minus that probability against the threshold. That comparison can never succeed at a threshold of a half or above, since one minus a probability of at least a half is at most a half, and the channel therefore proposes only where the cheaper route is also

the route the model prefers. Every operating point reported below sits at a half or above, the lowest exactly at it.

The threshold τ is selected on the tuning month for a target precision and reported unchanged on the evaluation months, and a proposal is counted as realised when the airline’s own subsequent filing chose the proposed route.

A flight is *eligible* under that policy whenever its two routes differ in planned fuel at all, which is 275 907 of the 281 720 held-out pairs, or 97.9 %, only exact ties being excluded. On those eligible flights the cheaper of the two routes is not usually the one the airline went on to file: in 55.1 % of them the fuel saver is the route the airline was in the act of abandoning, so the channel is arguing for staying put at least as often as for moving. A channel that proposed the cheaper route on every eligible flight would therefore be right 44.9 % of the time, which is worse than a coin toss.

That 44.9 % is the figure the model must exceed, and it is what makes the channel a learning problem rather than an arithmetic one. Finding the cheaper route is trivial; the whole task is telling the flights on which the cheaper route is the one the airline wants from the flights on which it is not. Everything that follows is the size of that gap.

That gap sits awkwardly beside Section 6.2, because the model reads a revision best when the airline was buying relief from delay, and the channel never proposes on those grounds: it proposes on fuel, which is the dimension along which route choice is hardest to call. The channel’s answer to that tension is not accuracy but selectivity, since it is free to say nothing, and on four eligible flights in five that is what it does; what carries it from 44.9 % to 79.2 % is the choice of when to speak, not a better reading of the flights it speaks about.

The simulation takes the airline’s own next filing as the alternative, which has three consequences, each taken up in full in Section 8: it bounds the evaluation to flights that did revise; it withholds the filing order the flight actually carried at the moment a channel would speak, so the channel must judge from the routes alone rather than from which one is already on file, which is why the 55.1 % of eligible pairs where the cheaper route is the one being abandoned are counted rather than set aside; and it means the fuel counted is not fuel the channel would create but fuel riding on a decision the model could identify in advance, a scale rather than a saving, since what a channel would genuinely add lies on the flights that never revise at all.

Precision is bought with volume, but not one for one: moving the target from 60 to 90 % raises achieved precision from 65.8 to 88.2 %, but cuts the

share of eligible flights spoken on to less than a third, while giving up 43 % of the confirmed fuel, 13.8 kt against 24.3 kt, because the proposals that survive the higher threshold are disproportionately the valuable ones.

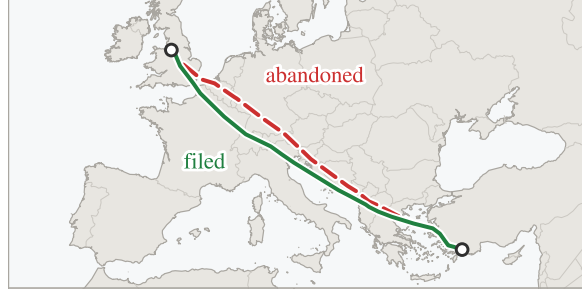
At the 80 % point used throughout this paper, the channel speaks on one eligible flight in five and surfaces 56 162 proposals over three months. The airlines' own subsequent filings confirm 79.2 % of them, against the 44.9 % that proposing on every eligible flight would have achieved, and the confirmed proposals carry 20.1 kt of planned fuel, equivalent to 63.5 kt of CO₂, at a mean of 452 kg per confirmed proposal.

Annualised, the confirmed fuel is 70 to 80 kt, or 222 to 254 kt of CO₂, the two ends being the two extrapolation bases available: the lower is the per-eligible-pair rate applied to the annual eligible volume estimated from the eighteen-month archive, and the upper is a simple factor of four on the quarter; the corresponding ranges at the other operating points follow the same pattern and reach 85 to 97 kt of fuel at the 60 % target. The median confirmed proposal saves 130 kg and the distribution is strongly skewed: the top tenth of confirmed proposals carries 64 % of the value.

7.2. Concentration of the value, and its sensitivity to the bias correction

If a channel of this kind were ever introduced, a small-scale trial could still reach most of the value, because the value is concentrated: of the 882 operators that receive any proposal, 6 fly half of the confirmed fuel and 28 fly four fifths of it, a pattern that holds more weakly by city pair (153 of the 10 939 pairs carry half the value) and is not the accident of a single month (the three test months contribute between 5.9 and 7.5 kt each). The counterweight is that the order in which operators would be reached is not neutral: the policy itself withholds nothing from any qualifying operator, but starting with six of them decides who benefits first.

Every fuel figure above uses the stay-pair-corrected model; repeating the whole simulation with the uncorrected one leaves the confirmed fuel unchanged to within fifty tonnes at either operating point tested ($\tau = 0.500$: 24.38 against 24.33 kt; $\tau = 0.600$: 20.101 against 20.097 kt), while precision moves by 1.4 to 2.3 points in whichever direction the threshold happens to favour (64.4 % against 65.8 % at $\tau = 0.500$; 81.5 % against 79.2 % at $\tau = 0.600$). That null result is expected rather than a disappointment: the correction targets flights that never revise, not the revised flights this simulation measures, so the headline fuel figures are robust to it even though the case for using the model at all rests on it.



	<i>abandoned</i>	<i>filed</i>
planned fuel [kg]	10 485	8 469
flight time [min]	259	258
route charges [EUR]	2 269	2 496
model score	-2.82	+0.90

$$\text{gap} = 3.7 \longrightarrow P = 0.99 > \tau = 0.600 \longrightarrow \text{proposed}$$

Figure 5: The worked example from the text, on an anonymised city pair. The route initially on file (dashed red) and the fuel-saving alternative (solid green) are shown. The alternative saves 2 016 kg of planned fuel at 227 EUR more in route charges and the same flight time. The calibrated acceptance probability exceeds the threshold; the airline’s own later filing chose the proposed route.

7.3. A worked example

One case from the held-out quarter, shown in Fig. 5, follows every step such a channel would take. A narrow-body was filed on a medium-haul flight with no regulation on either route; the alternative on offer burns 2 016 kg less planned fuel, close to a quarter of the trip fuel, at essentially the same flight time and 227 EUR more in route charges, a margin chosen for the figure because it is unambiguous and far from typical (the median confirmed proposal saves 130 kg). The model scores the alternative +3.7 ahead; the calibrator, fitted on the tuning month, maps that gap to a 99% acceptance probability, well above the threshold $\tau = 0.600$ fixed before the evaluation period began, and the channel proposes. The airline’s own later filing chose exactly that route: since the airline was told nothing, this is evidence not that a proposal would have been accepted, but that the model identified, before departure, a route the airline itself went on to prefer, which is the strongest validation available without running the channel and the form on which every fuel figure in this paper rests.

8. Threats to validity

The limitations of this study are collected here rather than scattered, in the order of how much they bound the claims.

The label is behaviour, not utility. A revision records what the airline’s filing process produced, folding together its economics, its dispatching software, its habits and its workload; the model learns this compound, arguably the right target for a proposal channel, since a proposal is accepted or ignored by the same compound. Two consequences follow. First, a revision need not be a choice at all: a route that ceased to be filable has to be replaced whatever the airline would have preferred, and the archive carries no field that separates such a revision from a deliberate one, so a share of the labels that cannot be quantified here records a constraint rather than a preference. Second, nothing in this paper measures a preference in the economic sense of Samuelson (1938) separately from the process that expresses it, and transfers of the learned function to settings where the process differs (e.g., another region’s filing culture) should be expected to degrade.

Flights that never revise. Every revision pair is a flight that changed its plan, yet a channel used in practice would see every flight, and most never revise, so the model would be asked a question on which it was evaluated only through the small stay set of Section 4.2. Precision should be expected to fall there, though not for the reason it is tempting to give: non-revision has several causes besides contentment with the filed plan (nothing prompted a second look, the dispatch desk had no capacity, the flight departed before a relevant regulation was published), and the channel’s premise, that the latter two are common, is not measured here. What the stay pairs do close, as Section 6.3 shows, is the model’s tendency to recommend change against traffic that stays, from 34.8 % to 90.0 %, and not by reading regulation status, since the model scores 88.0 % on the population where that shortcut would be wrong. Two bounds remain: the stay set is small (1 093 test pairs) and its two routes belong to different flights, so a model could learn the construction rather than the preference it encodes, the one place in this paper where a score compares routes across flights rather than within one. A balanced stay dataset, matching same-flight routes with regulation status independent of the label, requires data not currently assembled and is the single most valuable extension of this work. The never-revise limitation falls harder on the second use case of Section 2.2 than the first, since an overload release

ranks flights on the volume, most of which never revise, which is why that use case is described as enabled rather than demonstrated.

The candidates would come from somewhere else. In this simulation the alternative offered to a flight is the route the airline was about to file; a channel used in practice has no such route and would have to obtain candidates from a generator, precisely the distribution Section 2.3 argues is a poor source of *labels*. The argument transfers uncomfortably to serving time: a model trained on two routes that both passed an airline’s own flight-planning system may score a generated route low for reasons that have nothing to do with preference, a gap nothing in this paper measures, since every figure in Section 7 is conditioned on a candidate the airline had itself produced. A first trial should accordingly draw its candidates from routes that operator has filed before on that city pair, where the concern does not arise, rather than from a generator.

The route is carried as a vocabulary, not a geometry. The waypoint sequence enters the model as a bag of published identifiers, so the 1.4 points it is worth are knowledge of a particular vocabulary rather than of route shape, and two limits follow. Those identifiers are European, so a model fitted here would carry little of that knowledge to another region, which is the same non-transferability that the first of these limitations predicts for the filing process itself. They are also revised from time to time as the route network is republished, so a piece of airspace can reappear under a name the model has never met, with nothing in the encoding to connect the two; such changes are infrequent, and the effect is correspondingly slow, but it is a drift that only retraining repairs. Section 6.3 reports the attempt to replace the vocabulary with a learned representation of route shape, and why it did not succeed; a representation that does not depend on names would nonetheless be what makes the model portable, and remains the most valuable direction on this axis.

Fuel is not always a stable, burned cost. Two separate issues limit every fuel figure in this paper. First, every figure rests on planned fuel, which differs from burned fuel by whatever the tactical phase imposes (e.g., a direct routing granted en route, or a cruise level capped below the one requested); quantifying that difference requires flown trajectories matched to the plans that preceded them, and until that is done the totals of Section 7 should be read as an upper bound on what a proposal channel would deliver. Second,

planned fuel enters the model in kilogrammes and route charges in euros, two units that move on independent schedules: a kilogramme of jet fuel bought in 2025 is not the same expense as one bought in 2035, so any trade-off the model has learned between the two reflects the archive’s own prices rather than a stable preference, though nothing in eighteen months of training data teaches the model that this relationship drifts, because within that window it barely does. Recasting fuel as a monetary cost before training, or normalising every indicator to a common unit, would make the learned trade-offs portable across the years a model like this one would remain in use.

The counterfactual is absent. On every flight the simulation counts, the airline found the cheaper route by itself, so the 20.1 kt is not fuel that a channel would create but fuel that a channel could have surfaced earlier. What a channel would genuinely add, the flights on which the alternative was never considered, is precisely what this data cannot measure, and only a trial with real proposals and logged acceptance can close this gap and observe acceptance itself rather than its revealed proxy.

The regulation picture is the binding constraint at serving time. Attributed delay and the regulation picture are among the strongest inputs, and both are properties of a moment rather than of a flight, so a score computed against a regulation state that has since changed no longer describes the decision at hand: the binding constraint at serving time is data currency rather than computation, since the scoring step itself is a single pass through a fixed tree ensemble. Harder, and unmeasured here, is that the alternative route’s attributed delay is known in the archive only because the airline went on to file that route. For a route nobody has filed, the delay it would attract is a query rather than an observation, and the error in answering it would propagate into the acceptance probability.

9. Conclusions

Airline route acceptance is learnable from flight-plan revisions at network scale, and the findings of this paper are fourfold. First, a model fitted on 1 134 000 revisions identifies the chosen route in 68.1 % of held-out cases and supports operation at a selected precision rather than at a single accuracy figure: on the most confident fifth of the decisions it is right 94.9 % of the time. Secondly, no single cost indicator predicts the choice well on this archive, the strongest such rule reaching only 55.0 %. Thirdly, a proposal

channel tuned for 80 % precision and delivering 79.2 % addresses 20.1 kt of planned fuel per quarter, validated against what the airlines subsequently filed, four fifths of it flown by 28 of 882 operators, and these figures are robust to the stay-pair bias correction. Fourthly, route structure carries information about the choice that the cost indicators do not: the waypoint sequence is the largest single term in what moves the model’s score, though the cost indicators together outweigh it and removing the sequence costs only 1.4 points of accuracy.

A route that burns less planned fuel emits less carbon dioxide, and the reason such routes go unflown is not that nobody can find them but that nobody knows which of them an airline would actually take. That is the gap this paper addresses, and the reason acceptance rather than fuel is the quantity modelled: a proposal an airline declines saves nothing, and a channel that makes many of them is switched off. Ranking candidate routes by likely acceptance is what allows the environmentally cheaper proposals to be aimed at the flights on which they are also cheap for the operator, which is the only version of this idea that both sides have a reason to run.

The number that characterises this system is not 68.1 % but the curve of Fig. 3, because a channel chooses where on that curve to sit, and because ranking a pair correctly and proposing correctly are different quantities of which the second is the operational one: at the threshold used here, the channel proposes on about a fifth of the eligible flights, and 79.2 % of those proposals match what the airline went on to file. Were such a channel ever used, it would sit well to the left of that curve, moving right only as acceptance is observed. The second use case of Section 2.2, choosing which flights to ask to reroute when a volume of airspace is overloaded and in which order, remains enabled rather than demonstrated: it needs the same acceptance probability, on a population this archive cannot speak for.

Future work should assemble a balanced stay dataset in which regulation status is independent of the label, which is what currently bounds the change-bias measurement, and match flown trajectories to the plans that preceded them, so that planned savings can be restated as burned ones. Observing acceptance directly, rather than through the revealed proxy this paper relies on, remains an open question beyond its scope.

Only a live channel can reveal how preferences shift once proposals themselves begin to influence filing behaviour, so any practical use of this model would need to re-measure everything this paper measures.

Disclaimer

The views expressed are those of the authors and do not necessarily reflect the official position of EUROCONTROL or of its Member States. This paper reports research and does not commit to any operational service.

CRedit authorship contribution statement

Ramon Dalmau: Conceptualization, Methodology, Software, Data curation, Formal analysis, Investigation, Visualization, Writing – original draft. **David Perez:** Conceptualization, Resources, Validation, Writing – review & editing. **Franck Ballerini:** Validation, Writing – review & editing. **Seddik Belkoura:** Methodology, Validation, Writing – review & editing. **Vincent Taverniers:** Resources, Data curation, Writing – review & editing. **Gratiel Marius Marin:** Resources, Validation, Writing – review & editing. **Robin Deransy:** Conceptualization, Project administration, Writing – review & editing. **Benjamin Cramet:** Supervision, Writing – review & editing. **Grigori Gustin:** Supervision, Project administration, Writing – review & editing.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Claude (Anthropic) to assist with drafting and editing the manuscript text and with producing the analysis figures from the study’s result tables. Every analysis figure is drawn by the authors’ own code from the study’s result tables, and no figure in this paper was created or altered with generative AI. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data used in this study is operational air traffic data held by EUROCONTROL and processed under existing data-handling arrangements; it cannot be made publicly available. Aggregated tables underlying every figure are available from the corresponding author upon reasonable request, in the de-identified form this paper itself uses: operators, city pairs and waypoints appear in them by rank rather than by name, which leaves every number and every ordering unchanged. Tables holding one row per flight are not part of that offer.

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